Einstein's distant parallelism and Kron's machines

Does Einstein's 1928 teleparallel geometry answer Kron's non-Riemannian machine theory? — CLAUDE.md §3a, M9c.1

kronspice simulation report · 2026-11-07 · 120 V, 60 Hz motor, pushed to 40 rad/s, 4 s

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Summary

In 1928 Einstein built space-time from a **4-bein** — four frame legs at every point — with the rule that legs at different points are parallel ("distant parallelism"). That geometry has no curvature but has **torsion** Λ. He proposed that the trace of the torsion, φ_μ, is the electromagnetic potential, and that to first order the theory gives both vacuum gravity and Maxwell's equations. This example asks whether that answers Kron's multi-dimensional, non-Riemannian machine work, using Hoffmann's single-phase motor (see *hoffmann_motor*) as the test object.

- **One torsion, three notations:** Einstein's Λ equals minus the frame's anholonomy to 6×10^{-11}; Hoffmann's Ω equals Λ with one index lowered by the inductances to 3×10^{-17}. Kron's non-Riemannian frame is a teleparallel 4-bein.
- **Einstein's potential belongs to the frame, not the machine:** φ = 0 in Kron's frame (largest component 7×10^{-11}), but $\varphi_\theta = 0.300$ for the same motor in a frame whose rotor legs also grow as $e^{0.3\theta}$.
- **Curvature is the real difference:** Einstein's connection is flat (largest curvature component 0); the machine's own connection is curved (largest component 541).
- **Einstein's first approximation holds:** for plane waves his eqs. 7–9 hold to $\le 2\times10^{-16}$; on the light cone φ obeys Maxwell's vacuum equations.
- **A motor runs from the torsion alone:** built only from the 4-bein and its torsion, the kronspice motor reaches 376.8235 rad/s, equal to Kron's frame II(b) to 2×10^{-12} rad/s. With the torsion removed it only coasts to 26.8128 rad/s.

Background

A. Einstein, "New possibility for a unified field theory of gravitation and electricity," *Sitzungsber. Preuss. Akad. Wiss.* 1928, 224–227 (A. Unzicker's translation). Kron's non-Riemannian dynamics of rotating machines (1934), presented for physicists by Hoffmann (*Rev. Mod. Phys.* 21, 535, 1949), writes a machine's equations in non-holonomic frames — frames of current paths that are not the derivatives of any coordinates — and carries the motion in a torsion Ω (Hoffmann eq. 19). Hoffmann's motor has exactly four coordinates (three winding charges q1, q2, q3 and the rotor angle θ), so its change to stationary rotor coils (Hoffmann eq. 35) is a 4-bein, Einstein's case n = 4.

Standard position. Einstein's distant-parallelism unified theory was dropped by 1931; its gravity part survives as the teleparallel equivalent of general relativity, while electromagnetism from the torsion trace did not survive (Sauer 2006; Maluf 2013). This example is a comparison of formalisms, not a register hypothesis.

Results

1. Einstein's 4-bein is Kron's frame

the same torsion, computed as	agreement
Einstein's torsion Λ (his definition), in the frame's components	equals minus the frame's anholonomy (sign convention), to 6×10^{-11}
Hoffmann's Ω (*kron.Torsion*, his eq. 19)	equals Λ with one index lowered by the inductances, $\Omega_{\lambda\mu\nu} = L'_{\nu\rho}\cdot\Lambda^{\rho}{}_{\lambda\mu}$, to 3×10^{-17}
torsion of Einstein's own connection	2Λ, to 10^{-10}

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2. Einstein's electromagnetic potential on the motor

frame of the same motor	$\varphi_\mu = \Lambda^\alpha{}_{\mu\alpha}$ (trace of the torsion, his eq. 2)
Kron's stationary rotor coils (a rotation)	$(0, 0, 0, 7\times10^{-11})$, i.e. 0 — a “pure gravitational field” to Einstein
rotor legs also growing as $e^{0.3\theta}$	$\varphi_\theta = 0.300$

In a Kron machine the frame is a choice of current paths and both frames describe the same motor, so φ here is a property of the frame chosen, not a field the machine has. This echoes Einstein's own remark that “the separation of the gravitational and electromagnetic field seems arbitrary”.

3. Curvature: the real difference

connection	curvature
Einstein's distant parallelism, either frame	largest component 0 — flat; all its geometry is torsion
the machine's own (Christoffel symbols of its inductances and inertia)	largest component 541 (mixed units of henry and kg·m²) — not zero; the machine's space is curved

Kron's torsion is that curved space seen from a moving frame (Hoffmann eq. 17); a change of frame cannot remove curvature. Einstein's torsion *replaces* curvature. The mathematics of frames is the same; the geometry is different.

4. Einstein's first approximation, checked

His linearized field equation (eq. 5) was solved for plane waves $e^{p\cdot x}$, with x^4 imaginary as in his paper. $R_{\alpha\beta}$ is the Ricci tensor.

wave	solutions of eq. 5	largest φ	eq. 8	eq. 9	eq. 7	$R_{\alpha\beta}$ when φ = 0
$p^2 \neq 0$	4	6.3×10^{-17}	8×10^{-17}	9×10^{-17}	1×10^{-16}	2×10^{-16}
$p^2 \neq 0$, complex	4	4.2×10^{-16}	1×10^{-16}	5×10^{-17}	2×10^{-16}	3×10^{-16}
$p^2 = 0$ (light-like)	10	0.5	0	0	0	0

Away from the light cone the only solutions are coordinate changes (φ = 0, R = 0). On it, φ ≠ 0 obeys $\partial\varphi = 0$ and $\Box\varphi = 0$ — Maxwell's vacuum equations — and with φ = 0 the metric obeys $R_{\alpha\beta} = 0$. Einstein's first-approximation claims hold.

5. kronspice: the motor built from Einstein's torsion

The motor's machine model was built using only the 4-bein and its torsion: constant inductances L′ (Hoffmann's L-bar, the inductances in the stationary-coil frame) and a speed matrix $G = 2(\Omega_{\theta\lambda\nu} + \Omega_{\nu\lambda\theta})$, with no inductance derivatives. From the torsion G = [[0 0 0] [0 0 0.27] [−0.23 −0.23 0]], the same as *kron.RotatingFrame*. Run: 120 V, 60 Hz, pushed to 40 rad/s, 4 s.

model	final speed	ledger mismatch
frame I (coils turn with the rotor)	376.7928 rad/s	1×10^{-6}
Kron's frame II(b) (*kron.RotatingFrame*)	376.8235 rad/s	5×10^{-8}
built from Einstein's torsion	**376.8235 rad/s**	**5×10^{-8}**
same inductances, torsion removed	26.8128 rad/s — coasts on friction only	4×10^{-7}

The torsion-built motor and Kron's frame II(b) agree to 2×10^{-12} rad/s; frame I agrees to 0.03 rad/s (time step). Without the torsion there is no torque, and the rotor coasts as $40\cdot e^{-4\ s\cdot friction/J} = 26.81$ rad/s.

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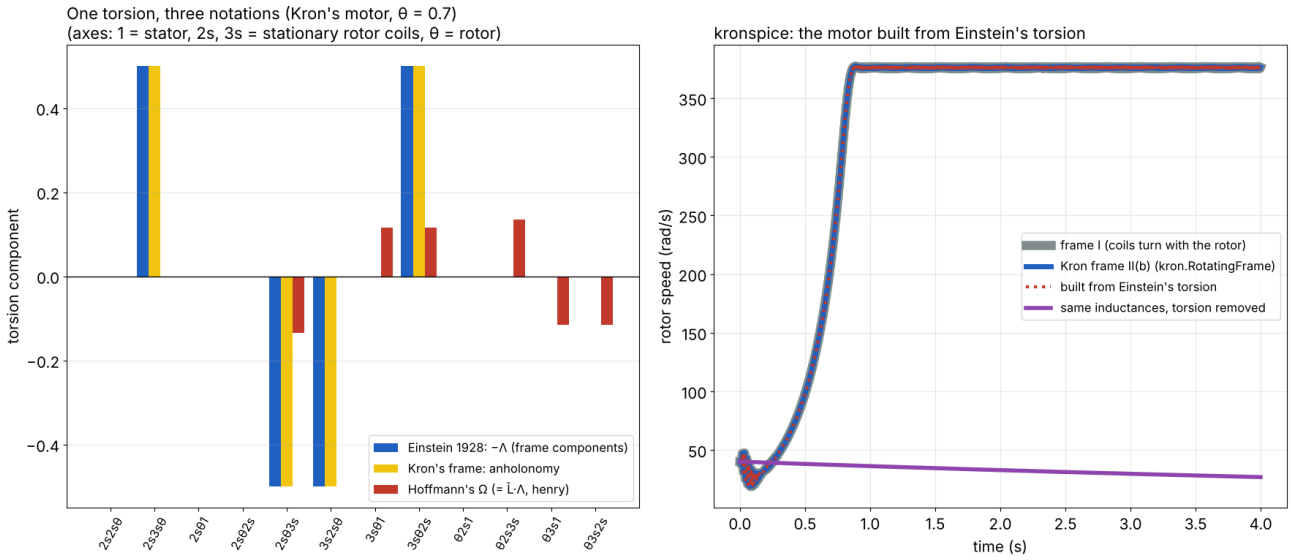


Figure 1. Left: the nonzero torsion components of Kron's motor at θ = 0.7 (axes: 1 = stator, 2s, 3s = stationary rotor coils, θ = rotor) in three notations — Einstein's −Λ in frame components (blue), the frame's anholonomy (yellow) and Hoffmann's Ω = L′·Λ in henry (red). Right: rotor speed of the four kronspice models; frame I, frame II(b) and the torsion-built motor overlap, while the motor without torsion only coasts down.

Validation

main_test.go checks the torsion three ways, φ and the curvature, the linear plane-wave check of eqs. 5–9 and $R_{\alpha\beta}$, and the motor runs. Every motor run closes its energy ledger to ≤ 1×10^{-6}.

Where it stands

- **Einstein's theory was dropped.** He worked on distant parallelism from summer 1928 to spring 1931 and gave it up because no single set of field equations could be justified (Sauer 2006).
- **Its gravity part survives** as the teleparallel equivalent of general relativity. Electromagnetism from the torsion trace did not survive.
- **For Kron.** Kron's non-Riemannian machine dynamics (1934; Hoffmann 1949) uses the same objects — n-beins, anholonomy and torsion — as a working engineering method. Its torsion is trace-free (φ = 0) and sits on a curved space, so it is not electromagnetism in Einstein's sense.
- **Not decided here:** whether Kron drew on Einstein's 1928–31 papers is a history question; it would take Kron's 1934 references to settle.

Reproducing the results

```
go run ./examples/einstein_teleparallel -viz examples/einstein_teleparallel/viz   # about 2 seconds
python3 examples/einstein_teleparallel/viz/render.py
go test ./examples/einstein_teleparallel
python examples/einstein_teleparallel/viz/make_report.py examples/einstein_teleparallel/einstein_teleparallel_report.pdf
```

Sources: A. Einstein, Sitzungsber. Preuss. Akad. Wiss. 1928, 217–221 and 224–227 (A. Unzicker's translation); B. Hoffmann, Rev. Mod. Phys. 21, 535 (1949); G. Kron, J. Math. Phys. 13, 103 (1934); T. Sauer, Historia Mathematica 33, 399 (2006); J. W. Maluf, Ann. Phys. 525, 339 (2013).

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